Stockholm University

Department of Meteorology

Modelling of Large-Scale Circulation in Atmosphere and Ocean (MO8004)

Wind Driven Ocean Gyres
Sverdrup and Munk Solutions

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Introduction

Ocean circulation gyres are part of the atmospheric circulation.

1. Heating air at the equator $\rightarrow$ Hadley cells.
2. Coriolis effect deflects the air movement $\rightarrow$ Trade winds and westerlies
3. Enforce wind stress on the ocean surface $\rightarrow$ 90°Ekman transport.
4. Since f increases with latitude, net southward Ekman transport in the North Atlantic.
5. Western boundary current needed to close the circulation.

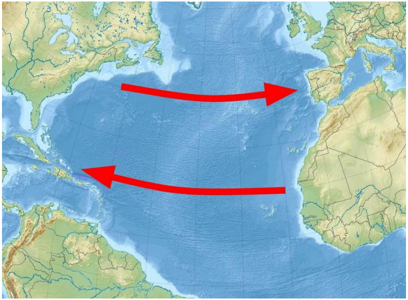

Figure: Wikimedia Commons, North Atlantic location map, accessed 2026

Introduction, Sverdrup

Harald Ulrik Sverdrup (1888-1957) formulated the net transport in the ocean interior as:

$$\boxed{-\beta\frac{\partial\Psi}{\partial x} = \frac{1}{\rho}\nabla\times\vec{\tau}} \tag{1}$$

which gives the Sverdrup solution:

$$\int_x^{x_{\text{East}}}\frac{\partial\Psi}{\partial x}\,dx = \int_x^{x_{\text{East}}} vH\,dx = \cancel{\Psi(x_{\text{East}},y)} - \Psi(x,y)$$

$$\implies \Psi(x,y) = -\frac{1}{\rho\beta}\int_x^{x_{\text{East}}}(\nabla\times\vec{\tau})$$

Introduction, Munk

Walter Heinrich Munk (1917-2019) extended the Sverdrup model by including meridional harmonic Newtonian viscosity (diffusion of momentum):

$$\boxed{A\nabla^4\Psi - \beta\frac{\partial \Psi}{\partial x} = \frac{1}{\rho}\nabla \times \vec{\tau}} \tag{2}$$

which gives the Munk solution:

$$\Psi(x,y) = \Psi_{\text{Sverdrup}}(x,y)\left(1 - e^{-\frac{x-x_{\text{West}}}{2\delta_m}}\left[\cos\left(\sqrt{3}\,\frac{x-x_{\text{West}}}{2\delta_m}\right) + \frac{1}{\sqrt{3}}\sin\left(\sqrt{3}\,\frac{x-x_{\text{West}}}{2\delta_m}\right)\right]\right)$$

$$\text{where } \delta_M \equiv \left(\frac{A}{\beta}\right)^{\frac{1}{3}}$$

Method

Stress set as:

$$\tau = \rho_{\text{air}} C_D U_{10}^2 \sin\left(\frac{\pi y}{L_y}\right). \tag{3}$$

The Sverdrup model:

- For the Atlantic Ocean: $L_x = L_y = 7000\,\text{km}$
- For the Pacific Ocean: $L_x = 14\,000\,\text{km}$ and $L_y = 7000\,\text{km}$.
- $\Delta x = \Delta y = 35\,\text{km}$
- $U_{10} = [4\,\text{m}\,\text{s}^{-1}, 5\,\text{m}\,\text{s}^{-1}, 6\,\text{m}\,\text{s}^{-1}]$

The Munk model:

- L_x, L_y, Δx ad Δy same as before.
- $U_{10} = 5\,\text{m}\,\text{s}^{-1}$
- $A = [3 \times 10^4\,\text{m}^2\,\text{s}^{-1}, 1 \times 10^5\,\text{m}^2\,\text{s}^{-1}, 5 \times 10^5\,\text{m}^2\,\text{s}^{-1}]$.
- $\implies \frac{\delta_M}{\Delta x} \approx [3.3, 4.9, 8.4]$

Result, The Sverdrup solution

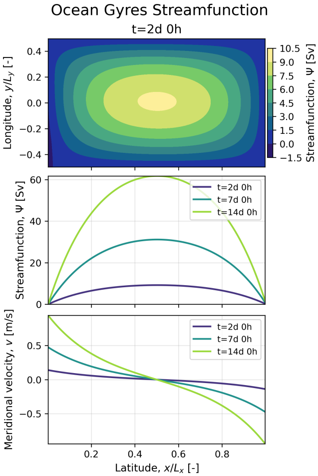

Figure: Streamfunction for the no-drag f-plane configuration. The circulation remains symmetric with no β-effect and without western intensification.

Result, The Sverdrup solution

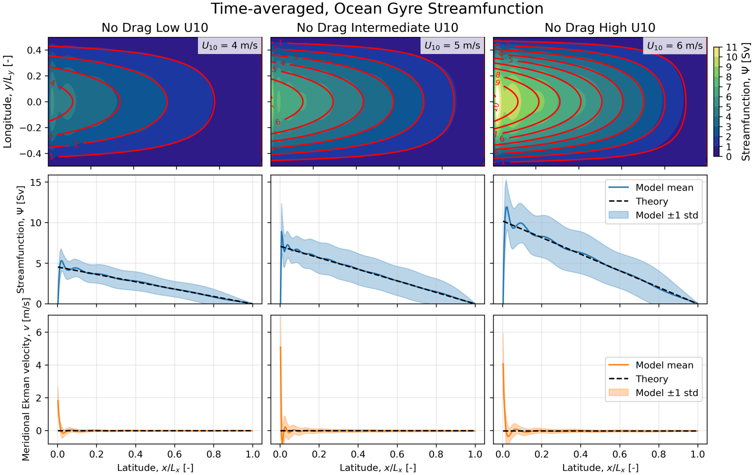

Figure: Streamfunction for the no-drag β-plane configuration. The red lines are the theoretical Sverdrup solution.

Result, The Munk solution

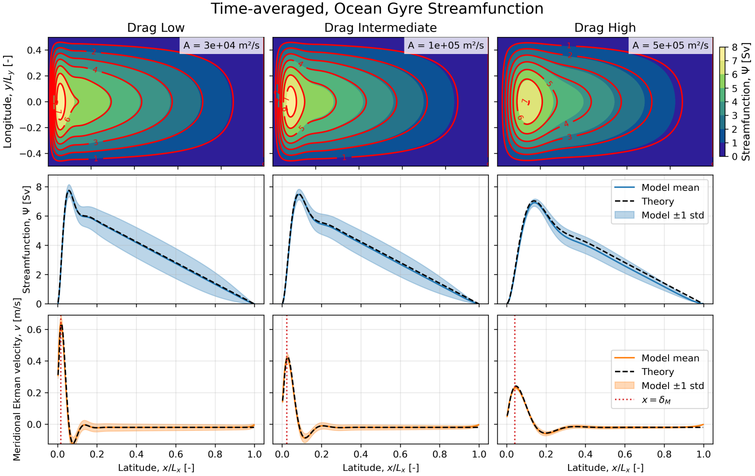

Figure: Streamfunction for the Laplace-viscous β-plane configuration. The red lines are the theoretical Munk solution.

Result, The Munk solution

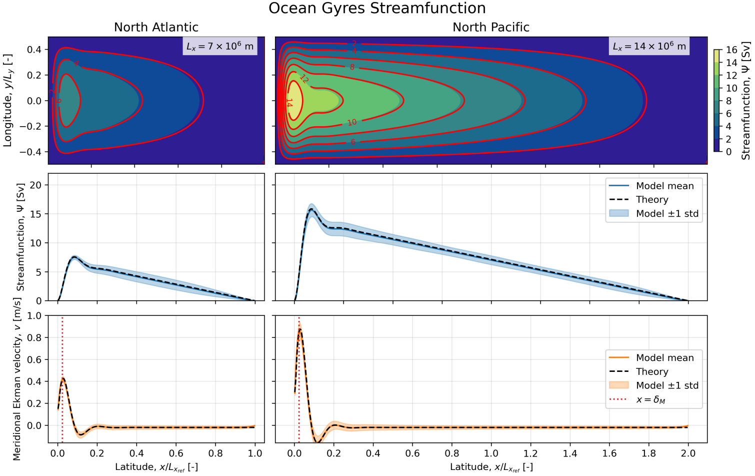

Figure: Streamfunction for the drag β-plane configuration with varying domain size. The red lines are the theoretical Sverdrup solution.

Result, The Munk solution

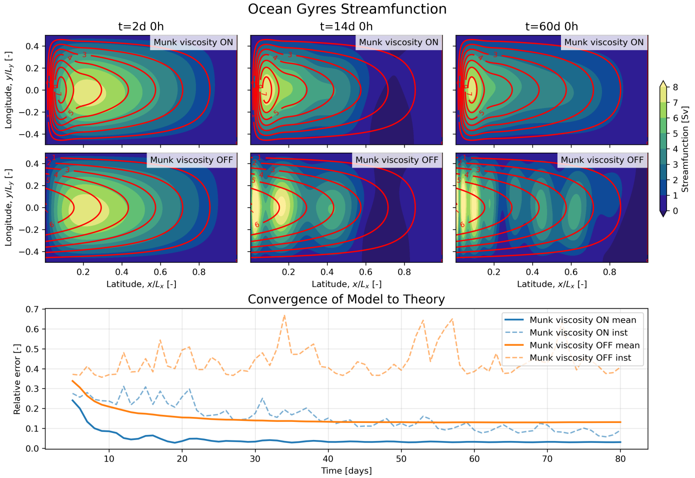

Figure: Streamfunction for the Laplace-viscous β-plane configuration. The red lines are the theoretical Sverdrup and Munk solutions respectively.

Discussion

Both numerical models show western boundary currents, but only Munk theory does. Why?

- The Sverdrup theory is linear, whereas the Munk theory is non-linear, and frictional force is opposite sign to the motion itself.
- The Munk viscosity creates a boundary layer where the frictional force can take care of mass conservation.
- Friction balances continuos input of energy. *Realistic*?
- Gives western boundary flow in right magnitude to real Gulf Stream ($\sim 2.5\,\mathrm{m\,s^{-1}}$).

$$A\nabla^4\Psi - \beta\frac{\partial\Psi}{\partial x} = \frac{1}{\rho}\nabla \times \vec{\tau}$$

Conclusion

Three important conclusions:

1. Both numerical models showed western intensification, explained by Rossby wave propagation. But continuous energy input need dissipation to converge with time.
2. Only Munk theory showed a western boundary current, explained by frictional layer, although both numerical models showed return flow.
3. A larger domain gave same interior meridional velocity, which resulted in a stronger western boundary current.

References

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